

# Applications: Growth and Decay



## Growth models

- 1 A city's population is modeled by  $P(t) = 12000(1.035)^t$ , where  $t$  is in years.
    - a State the initial population and the annual growth rate.
    - b Find the population after 10 years, to the nearest person.
  - 2 An account earns 4% annual interest compounded continuously. How long does it take an investment to double? Give an exact expression and a decimal to two places.
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## Decay models

- 3 Carbon-14 has a half-life of 5730 years:  $A(t) = A_0\left(\frac{1}{2}\right)^{t/5730}$ .
    - a What fraction of the original carbon-14 remains after 2000 years? Answer to three decimal places.
    - b A sample retains 30% of its carbon-14. Estimate its age to the nearest hundred years.
  - 4 A 200 g sample of a radioactive substance decays to 50 g in 12 hours, following  $A(t) = 200e^{kt}$ .
    - a Find the decay constant  $k$  (4 d.p.).
    - b Find the half-life of the substance.
  - 5 Newton's law of cooling: coffee at  $90^\circ\text{C}$  cools in a  $20^\circ\text{C}$  room following  $T(t) = 20 + 70e^{-0.1t}$  ( $t$  in minutes). Find the temperature after 10 minutes, to one decimal place.
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## Logistic growth (an introduction)

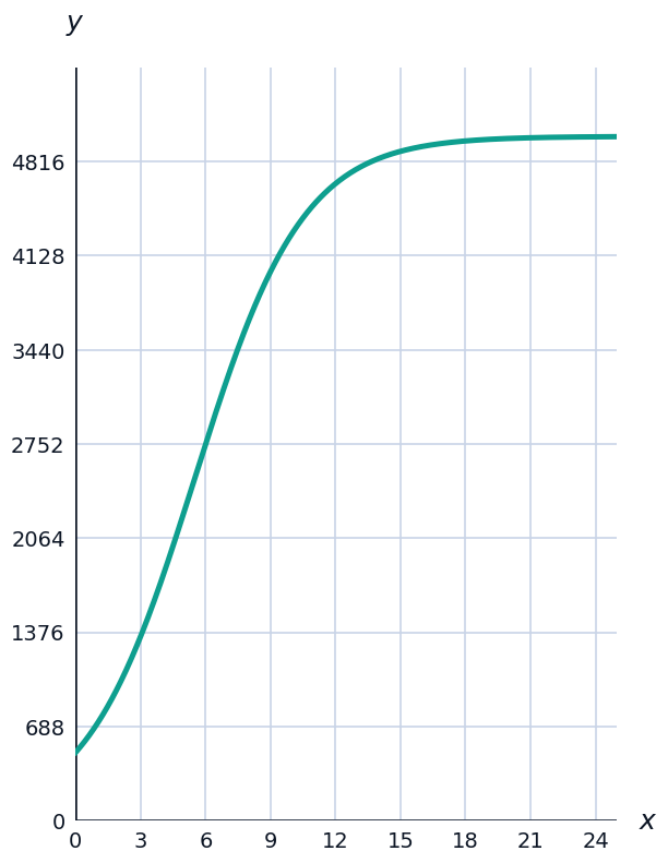
- 6 A lake is stocked with fish, and the population is modeled by  $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$ , where  $t$  is in years.
    - a Find the initial population  $P(0)$ .
    - b Find the carrying capacity (the value  $P$  approaches as  $t \rightarrow \infty$ ).
    - c Find  $P(10)$ , to the nearest fish.
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## Comparing models

- 7 Two towns start with 10,000 people. Town A grows at a constant rate of 300 people/year (linear); Town B grows at 3% per year (exponential). Find each town's population after 15 years, to the nearest person, and state which is larger.
- 8 A radioactive isotope decays from 80 g to 10 g over 3 half-lives. Find the half-life in terms of the decay time, and write the exponential decay model  $A(t) = 80e^{kt}$ , finding  $k$  if the total elapsed time is 15 years (4 d.p.).

### Visualizing the logistic curve

- 9 The fish population model  $P(t) = \frac{5000}{1 + 9e^{-0.4t}}$  (from earlier) is graphed below. Use the graph to estimate where the population is growing fastest, then explain why this happens near the inflection point.



### Doubling time and half-life formulas

- 10 Derive a general formula for the doubling time  $T$  of a quantity growing as  $A(t) = A_0e^{kt}$ , in terms of  $k$ .
- 11 Using the doubling-time formula derived above, find the doubling time for a population growing at a continuous rate of  $k = 0.025$  per year, to one decimal place.

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### Half-life formula derivation

- 12** Derive a general formula for the half-life  $T_{1/2}$  of a quantity decaying as  $A(t) = A_0 e^{kt}$  (with  $k < 0$ ), in terms of  $k$ .
- 13** A radioactive isotope has decay constant  $k = -0.023$  per year. Use the formula above to find its half-life, to one decimal place.

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### Combining growth models with rates

- 14** A savings account earns continuous interest, growing from 5000 to 5512.50 riyals in 2 years. Find the annual interest rate  $r$  using  $A = Pe^{rt}$ , to two decimal places (as a percentage).